

Modeling and Evaluation of Spaceborne SAR Onboard Ship Detection Response to Midterm Review Comments

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1. Equivalence of Eq. (1) and Eq. (6)

The data fidelity term in Eq. (1) is defined in the raw echo domain ($\|Y - A(H)\|^2$), whereas the one in Eq. (6) is defined in the image domain ($\|H - H_{\text{CSA}}\|^2$). What is the theoretical basis for treating the two as equivalent?

Response:

The data fidelity terms in Eq. (1) and Eq. (6) are equal under the forward operator used in this study, as shown below.

Following Bi *et al.* [6] in the report, $\mathcal{A}$ is taken as the inverse of the CSA imaging operator $\mathcal{I}$, i.e., $\mathcal{A} = \mathcal{I}^{-1}$. Let $\mathbf{y} = \text{vec}(\mathbf{Y})$ and $\mathbf{h} = \text{vec}(\mathbf{H})$ be the vectorized echo and image. Since CSA is linear, it can be written as a matrix Φ , so that

$$\mathbf{h}_{\text{CSA}} = \Phi \mathbf{y}, \quad \text{vec}(\mathcal{A}(\mathbf{H})) = \Phi^{-1} \mathbf{h}.$$

The key property is that Φ is unitary, i.e., $\Phi^H \Phi = \Phi \Phi^H = \mathbf{I}$, and hence $\Phi^{-1} = \Phi^H$. Please refer to Appendix A for details of the vectorization, the matrix form of Φ , and the proof of its unitarity. With this property, the data fidelity term of Eq. (1) becomes

$$\begin{aligned} \|\mathbf{Y} - \mathcal{A}(\mathbf{H})\|_F^2 &= \|\mathbf{y} - \Phi^{-1} \mathbf{h}\|_2^2 && \text{(vectorization preserves the norm)} \\ &= \|\Phi^{-1} (\mathbf{h}_{\text{CSA}} - \mathbf{h})\|_2^2 && (\mathbf{y} = \Phi^{-1} \mathbf{h}_{\text{CSA}}) \\ &= \|\Phi^H (\mathbf{h}_{\text{CSA}} - \mathbf{h})\|_2^2 && (\Phi^{-1} = \Phi^H) \\ &= (\mathbf{h}_{\text{CSA}} - \mathbf{h})^H \Phi \Phi^H (\mathbf{h}_{\text{CSA}} - \mathbf{h}) && (\|\mathbf{v}\|_2^2 = \mathbf{v}^H \mathbf{v}) \\ &= (\mathbf{h}_{\text{CSA}} - \mathbf{h})^H (\mathbf{h}_{\text{CSA}} - \mathbf{h}) && (\Phi \Phi^H = \mathbf{I}) \\ &= \|\mathbf{h}_{\text{CSA}} - \mathbf{h}\|_2^2 = \|\mathbf{H}_{\text{CSA}} - \mathbf{H}\|_F^2. \end{aligned}$$

The two data fidelity terms are therefore equal, so Eq. (1) with $R(\mathbf{H}) = \|\mathbf{H}\|_1$ and Eq. (6) are the same problem. In the image-domain form of Eq. (6), the problem decouples into independent per-pixel problems, which gives the element-wise closed-form solution in Eq. (7). Invertibility alone is not sufficient: if Φ were not unitary, the fidelity term would become

the weighted norm $\|\Phi^{-1}(\mathbf{h}_{\text{CSA}} - \mathbf{h})\|_2^2$, which couples the pixels and has no element-wise closed-form solution.

2. Is CS Undersampling Actually Applied?

Sections 1 and 2 state that CS can reconstruct the image from fewer measurements than the Nyquist rate requires. However, according to Table 3, $f_s = 64$ MHz exceeds $B = 50$ MHz in range, and $\text{PRF} = 1$ kHz exceeds the Doppler bandwidth of about 200 Hz in azimuth, so the echo Y is fully sampled in both directions. Is the echo undersampled at any point in this study? If not, what role does CS play in the overall pipeline? Are the images fed into YOLO obtained from a reduced number of measurements?

Response:

As shown in [6] in the report, sparse SAR imaging can either reconstruct the image from undersampled echo, or make the image clearer from fully sampled echo. This study adopts the latter: no undersampling is applied, CS serves as the sparse regularization that enhances the CSA complex image, and the images fed into YOLO come from the fully sampled echo. We will remove the descriptions of sub-Nyquist reconstruction to avoid misunderstanding.

Fig. 1 shows this effect on a single point target. After refinement, the mainlobe extent decreases from 2.50 to 2.25 pixels in range and from 2.62 to 2.25 pixels in azimuth, and all sidelobes are removed except the tips of the first sidelobes. Setting λ to the peak sidelobe level (-13.3 dB) may perform better, removing all sidelobes, but may also suppress weak targets. Since thresholding sees only pixel intensity, it also cannot tell a sidelobe from overlapping sinc responses of nearby targets, so it is expected to help mainly isolated targets or object edges in sparse scenes; both points will be evaluated in future work.

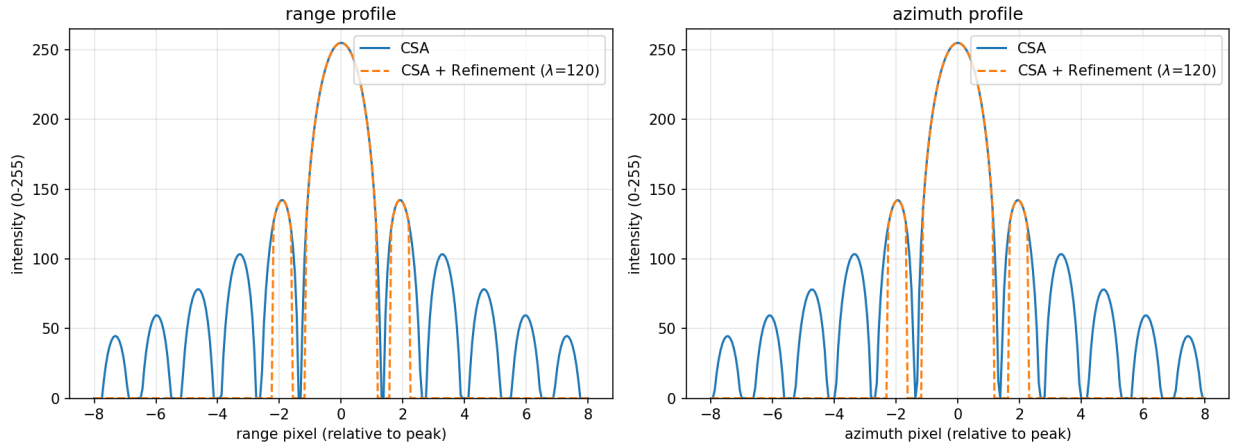


Figure 1: Range (left) and azimuth (right) profiles of a single point target (0–255 intensity, 30 dB dynamic range) from CSA (solid) and CSA + refinement with $\lambda = 120$ (dashed). Pixels below λ are set to zero, which is equivalent to Eq. (7) with λ added back.

3. Computational Complexity Analysis

Sections 1 and 2 state that CS can reduce the computational load, which supports real-time onboard processing. However, the one-step refinement is an additional step after CSA.

Shouldn't the computational load therefore be higher than CSA alone? What baseline is the "reduced computation" measured against? What is the approximate computational load or execution time of the whole pipeline (CSA, refinement, and YOLO)?

Response:

The claim of low computational complexity in Sections 2.2 and 2.3 is made relative to transformer-based methods, not to CSA alone. The requirement of onboard processing is to achieve better performance than conventional CSA without adding much computational complexity. A quantitative analysis of the computational load of the whole pipeline (CSA, refinement, and YOLO) will be conducted in future work.

4. Symbol Definitions and Dimensions

Fig. 3 uses H to generate Y , while Eq. (4) uses $S = [s_{m,n}]$ to denote the echo. What is the relationship between Y and S ? In addition, most symbols in the report are given without their dimensions or data types. For example, is H a matrix? What is its dimension? Is it complex-valued, or real-valued intensity in the range 0–255? Do other symbols have the same issue?

Response:

The received echo is $\mathbf{Y} = \mathbf{S} + \mathbf{W}$, where $\mathbf{S} = [s_{m,n}]$ in Eq. (4) is the noise-free echo synthesized from the scene $\mathbf{H}$, and $\mathbf{W}$ is the thermal noise, modeled as i.i.d. circular complex Gaussian. In the current experiments, $\mathbf{W}$ is set to zero, i.e., $\mathbf{Y} = \mathbf{S}$. $\mathbf{H}$ is a complex-valued $M \times N$ matrix. For detection, the CSA complex image is cropped to 800×800 pixels and converted to a real-valued 8-bit intensity image (30 dB dynamic range mapped to 0–255), on which the threshold $\lambda = 120$ is applied. The main symbols in the report, with their dimensions and data types, are listed in Table 1; we will add this information to the report.

Table 1: Main symbols in the report, with their dimensions and data types.

Symbol	Description	Type / dimension
<i>Echo and image</i>		
M, N	Numbers of azimuth and range samples	Positive integers
m, n	Azimuth and range sample indices	Integers
$\mathbf{Y}$	Received echo	$\mathbb{C}^{M \times N}$
$\mathbf{S} = [s_{m,n}]$	Noise-free echo, Eq. (4)	$\mathbb{C}^{M \times N}$
$\mathbf{W}$	Thermal noise	$\mathbb{C}^{M \times N}$
$\mathbf{H}$	Scene reflectivity	$\mathbb{C}^{M \times N}$
$\mathbf{H}_{\text{CSA}}$	CSA complex image	$\mathbb{C}^{M \times N}$
$\hat{\mathbf{H}}$	Refined complex image, Eq. (6)–(7)	$\mathbb{C}^{M \times N}$
–	Intensity image fed to YOLO	$\{0, \dots, 255\}^{800 \times 800}$
<i>Operators and regularization</i>		
$\mathcal{A}(\cdot)$	Forward operator, Eq. (1)	$\mathbb{C}^{M \times N} \rightarrow \mathbb{C}^{M \times N}$
$R(\cdot)$	Sparsity regularizer, Eq. (1)	$\mathbb{C}^{M \times N} \rightarrow \mathbb{R}_{\geq 0}$
λ (Eq. (1), (6)–(7))	Regularization parameter / threshold	Real scalar, ≥ 0
$S_{\lambda}(\cdot)$	Soft-thresholding, Eq. (7)	$\mathbb{C} \rightarrow \mathbb{C}$, element-wise

Appendix A: Unitarity of the CSA Matrix

This appendix supports the response to Comment 1 by showing that the CSA imaging operator, written as a matrix Φ , is unitary.

A.1 Vectorization

The echo $\mathbf{Y}$ and the image $\mathbf{H}$ are both complex-valued $M \times N$ matrices, where M and N are the numbers of azimuth and range samples. The operator $\text{vec}(\cdot)$ stacks the columns of a matrix into one long column vector:

$$\mathbf{y} = \text{vec}(\mathbf{Y}) = \begin{bmatrix} \mathbf{Y}_{:,1} \\ \vdots \\ \mathbf{Y}_{:,N} \end{bmatrix} \in \mathbb{C}^{MN}, \quad \mathbf{h} = \text{vec}(\mathbf{H}) \in \mathbb{C}^{MN},$$

where $\mathbf{Y}_{:,n}$ is the n -th column of $\mathbf{Y}$. Vectorization only rearranges the entries, so it does not change the norm: $\|\mathbf{Y}\|_F = \|\mathbf{y}\|_2$. It also turns every 2-D operation in CSA into a matrix-vector product, which allows the whole algorithm to be written as a single matrix.

A.2 CSA Flow in Matrix Form

CSA consists of the following steps: azimuth FFT, chirp scaling, range FFT, range compression and bulk RCMC, range IFFT, azimuth compression and residual phase compensation, and azimuth IFFT. In matrix form,

$$\Phi = \mathbf{F}_a^{-1} \mathbf{P}_3 \mathbf{F}_r^{-1} \mathbf{P}_2 \mathbf{F}_r \mathbf{P}_1 \mathbf{F}_a,$$

where

- $\mathbf{F}_a = \mathbf{I}_N \otimes \mathbf{F}_M$ is the azimuth FFT (applied within each column) and $\mathbf{F}_r = \mathbf{F}_N \otimes \mathbf{I}_M$ is the range FFT (applied within each row). Here $\otimes$ is the Kronecker product and $\mathbf{F}_K$ is the K -point DFT matrix.
- $\mathbf{P}_1$ is the chirp scaling, $\mathbf{P}_2$ is the range compression and bulk RCMC, and $\mathbf{P}_3$ is the azimuth compression and residual phase compensation.

Since a product of unitary matrices is unitary, it suffices to show that each factor is unitary.

A.3 Chirp Scaling and Phase Compensation

Chirp scaling and phase compensation are both Hadamard (element-wise) products of the data with a complex exponential array. For chirp scaling in the range-Doppler domain,

$$\mathbf{X}_{\text{out}} = \mathbf{E}_1 \odot \mathbf{X}_{\text{in}}, \quad [\mathbf{E}_1]_{m,n} = e^{j\phi_1(f_{a,m}, \tau_n)},$$

where ϕ_1 is real. After vectorization, the Hadamard product becomes multiplication by a diagonal complex exponential matrix:

$$\text{vec}(\mathbf{X}_{\text{out}}) = \mathbf{P}_1 \text{vec}(\mathbf{X}_{\text{in}}), \quad \mathbf{P}_1 = \text{diag}(\text{vec}(\mathbf{E}_1)) = \text{diag}(e^{j\phi_1}, \dots, e^{j\phi_1, MN}),$$

with MN diagonal entries. Since every diagonal entry has unit magnitude,

$$\mathbf{P}_1^H \mathbf{P}_1 = \text{diag}(|e^{j\phi_{1,1}}|^2, \dots, |e^{j\phi_{1,MN}}|^2) = \mathbf{I},$$

so $\mathbf{P}_1$ is unitary. The same holds for $\mathbf{P}_2$ and $\mathbf{P}_3$, which differ from $\mathbf{P}_1$ only in the phase function.

A.4 DFT

The normalized DFT matrix $\mathbf{F}_K$ is unitary, and the Kronecker product of unitary matrices is unitary, so $\mathbf{F}_a$ and $\mathbf{F}_r$ are unitary. Our implementation uses an unnormalized forward FFT (a factor of $\sqrt{K}$) and an inverse FFT divided by K (a factor of $1/\sqrt{K}$). Since each axis has exactly one FFT and one inverse FFT, the two factors cancel.

A.5 Conclusion

All factors of Φ are unitary, so Φ is unitary: $\Phi^H \Phi = \mathbf{I}$ and $\Phi^{-1} = \Phi^H$.